

UEL Product Data Sheet Notes

Clean notes version with product details, applications, features, performance claims, and typical technical values

Source note: This document is prepared from the uploaded UEL product data sheets. All technical values are listed as typical values and may vary, following the wording shown in the data sheets.

Important note: AWS68 is described as an anti-wear hydraulic oil, while its application section states gearbox/final drive/differential use. This document keeps the application exactly as stated in the data sheet.

1. Overall Product Summary

No.	Product	Product Type
1	AWS32	Premium anti-wear hydraulic oil
2	AWS68	Premium anti-wear hydraulic oil
3	Cutting Fluid Coolant	Cutting fluid coolant
4	5W30 SQ/CF Fully	Fully synthetic engine oil
5	5W30 SQ/CF Semi	Semi-synthetic engine oil
6	10W30 SL/CF Mineral	Mineral engine oil
7	10W40 SQ/CF Semi	Semi-synthetic engine oil
8	15W40 Fully	Fully synthetic engine oil
9	15W40 SM/CF Mineral	Mineral engine oil
10	15W40 SQ/CF Semi	Semi-synthetic engine oil
11	20W50 SL/CF Mineral	Mineral engine oil
12	ATF DEXRON-III	Automatic transmission fluid
13	HD 40	Monograde diesel engine oil

2. Product Notes

1. AWS32

Product type: Premium anti-wear hydraulic oil

Description: Premium anti-wear hydraulic oil made from highly refined base oils and zinc-based additives. It provides protection against wear, oxidation, and corrosion, with reliable performance and long service life in demanding hydraulic systems.

Application: Used in industrial and mobile hydraulic systems, including forklifts, excavators, injection molding machines, compressors, and factory machinery. It helps provide smooth operation, reduce wear, prevent corrosion, and maintain stable performance under high pressure.

Product performance claim: ISO VG 32; DIN 51524 Part II (HLP anti-wear hydraulic fluids); AW / AWS Anti-Wear Hydraulic Oil Standards.

Features and advantages:

- Excellent wear protection for longer equipment life
- High oxidation stability keeps systems clean
- Good rust and corrosion resistance
- Stable viscosity and reliable performance under pressure

Typical technical results:

Parameter	Method	Typical Result
Viscosity Grade	SAE J300	AWS32
Colour	VISUAL	Visual
Density @ 15°C	ASTM D4052	861.8 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	32.12 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	5.9071 mm ² /s
Viscosity Index	ASTM D2270	112.4
Pour Point	ASTM D97	-33°C
Flash Point	IP 523	210°C

2. AWS68

Product type: Premium anti-wear hydraulic oil

Description: AWS68 is a premium anti-wear hydraulic oil made from highly refined Group II base oils and zinc-based additives. It provides protection against wear, oxidation, and corrosion. The clean “white oil” formulation supports reliable performance and long service life in demanding hydraulic systems.

Application: The data sheet states it is recommended for use in automotive gearboxes, final drives, and differentials where API GL-4 or GL-5 performance is required. It is suitable for trucks, buses, tractors, and off-road equipment operating under severe or high-temperature conditions.

Product performance claim: ISO VG 68; DIN 51524 Part II (HLP anti-wear hydraulic fluids); AW / AWS Anti-Wear Hydraulic Oil Standards.

Features and advantages:

- Excellent wear protection for longer equipment life
- High oxidation stability keeps systems clean
- Good rust and corrosion resistance
- Fast water separation and easy filtration
- Stable viscosity and reliable performance under pressure

Typical technical results:

Parameter	Method	Typical Result
Viscosity Grade	SAE J300	AWS68
Colour	VISUAL	Visual
Density @ 15°C	ASTM D4052	863.6 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	64.694 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	9.1529 mm ² /s
Viscosity Index	ASTM D2270	126.46
Pour Point	ASTM D97	-18°C
Flash Point	IP 523	220°C

3. Cutting Fluid Coolant

Product type: Cutting fluid coolant

Description: Cutting Fluid provides cooling and lubrication during metal cutting and machining operations. It reduces friction, prevents tool wear, supports smooth surface finishes, provides rust protection, and helps improve productivity and precision.

Application: Suitable for drilling, milling, turning, grinding, and tapping operations on ferrous and non-ferrous metals. It is suitable for CNC machines, lathes, and general machining tools to support smoother cuts, extended tool life, and improved surface quality.

Product performance claim: Not stated in the product data sheet.

Features and advantages:

- Excellent cooling and lubrication performance
- Reduces friction and tool wear
- Improves surface finish and machining accuracy
- Provides strong rust and corrosion protection
- Extends tool and fluid life for higher productivity

Typical technical results:

Parameter	Method	Typical Result
Appearance	-	Neat yellow color liquid
Appearance after dilution	-	White translucent
Odour	-	Mild
Density @ 15°C	-	1.02
pH value (5.0% Emulsion)	-	9.6
Solubility	-	Water
Synthetic Oil / Mineral Oil	-	15 to 30%
Fatty Oil (Vegetable type)	-	1.0 to 5.0%
Anti-corrosion additives	-	2.0 to 5.0%
Brix reading factor	-	1.11

4. 5W30 SQ/CF Fully

Product type: Fully synthetic engine oil

Description: A high-performance fully synthetic engine oil formulated with advanced synthetic base stocks and a superior additive system. It delivers outstanding lubrication, wear protection, and thermal stability for gasoline and light-duty diesel engines. It also reduces deposits and supports smooth, efficient engine performance across a wide range of temperatures.

Application: Recommended for gasoline and light-duty diesel engines in passenger cars, SUVs, light trucks, and commercial vehicles. Ideal for high-temperature environments, frequent stop-start conditions, long-distance driving, and modern vehicles requiring stable, high-performance fully synthetic protection.

Product performance claim: API SQ/CF; GM dexos1TM Gen 2; Ford WSS-M2C946-A / WSS-M2C946-B1; Chrysler MS-6395; MB-Approval 229.5 / 229.51; VW 502.00 / 505.00; BMW Longlife-01.

Features and advantages:

- Superior protection against wear, sludge, and deposits
- Excellent cold-start performance with fast lubrication
- Outstanding oxidation and high-temperature stability
- Reduced friction for smoother and more efficient engine operation
- Suitable for gasoline (SQ) and older diesel engines (CF)
- Consistent performance in city, highway, and heavy-use conditions

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	853.9 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	78.85 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	11.743 mm ² /s
Viscosity Index	ASTM D7042	159.35
Pour Point	ASTM D97	-35°C

Flash Point	IP 523	220°C
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5. 5W30 SQ/CF Semi

Product type: Semi-synthetic engine oil

Description: A high-performance engine oil formulated with a blend of synthetic and refined mineral base oils. It provides enhanced lubrication, excellent wear protection, and strong thermal stability for gasoline and light-duty diesel engines. The formulation helps reduce friction, control deposits, and maintain smooth operation across different temperatures and driving conditions.

Application: Suitable for gasoline and light-duty diesel engines used in passenger cars, SUVs, light trucks, and commercial vehicles. Recommended for urban driving, frequent start-stop use, long-distance travel, and moderate to high-temperature environments where improved fuel efficiency and stable semi-synthetic performance are required.

Product performance claim: API SQ/CF; GM dexos1TM Gen 2; Ford WSS-M2C946-A / WSS-M2C946-B1; Chrysler MS-6395; MB-Approval 229.5 / 229.51; VW 502.00 / 505.00; BMW Longlife-01.

Features and advantages:

- Enhanced wear, sludge, and deposit protection
- Excellent cold-start performance due to low 5W viscosity
- Improved oxidation and thermal stability
- Reduced friction for smoother and more efficient engine operation
- Suitable for gasoline (SQ) and older diesel engines (CF)
- Reliable performance for daily driving, long trips, and mixed driving conditions

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	865.4 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	94.530 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	12.893 mm ² /s
Viscosity Index	ASTM D2270	133.45
Pour Point	ASTM D97	-27°C
Flash Point	IP 523	220°C

6. 10W30 SL/CF Mineral

Product type: Mineral engine oil

Description: A high-quality mineral engine oil formulated with refined base oils and a balanced additive system. It provides reliable lubrication, wear protection, and thermal stability for gasoline and light-duty diesel engines. The formulation helps minimize deposits, reduce friction, and support smooth engine operation under different driving conditions.

Application: Suitable for gasoline and light-duty diesel engines in passenger cars, light trucks, small commercial vehicles, and general-purpose equipment. Recommended for moderate climates, mixed driving conditions, and daily commuting where consistent mineral-based protection is required.

Product performance claim: API SL/SJ/CF; ACEA A3/B3 or A3/B4.

Features and advantages:

- Strong protection against wear, sludge, and deposits
- Stable viscosity and good lubrication at operating temperatures

- Reduced friction for smoother and more efficient engine performance
- Suitable for gasoline (SL) and older diesel engines (CF)
- Reliable performance for urban, highway, and light commercial use

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	867.3 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	88.01 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	10.229 mm ² /s
Viscosity Index	ASTM D2270	119.01
Pour Point	ASTM D97	-11°C
Flash Point	IP 523	240°C

7. 10W40 SQ/CF Semi

Product type: Semi-synthetic engine oil

Description: A high-quality semi-synthetic engine oil designed to provide excellent protection and performance for a wide range of gasoline and diesel engines. It is formulated to ensure superior lubrication under high temperature and heavy-load conditions, reducing wear and extending engine life.

Application: Suitable for modern passenger cars, light commercial vehicles, and motorcycles. Recommended for engines operating in city traffic or long-distance driving conditions, ensuring stable performance and fuel efficiency.

Product performance claim: API SQ/CF-4; BMW Longlife-01; Chrysler MS-6395; Fiat 9.55535-G2; Mercedes-Benz MB 229.1 / 229.3; PSA B71 2294; Renault RN 0700 / RN 0710.

Features and advantages:

- Excellent high-temperature protection and oxidation stability
- Reduces engine wear and deposits
- Enhances fuel efficiency
- Compatible with a wide range of engine types
- Meets API SQ/CF performance standards

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	870.6 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	148.36 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	16.21 mm ² /s
Viscosity Index	ASTM D2270	143.91
Pour Point	ASTM D97	-26°C
Flash Point	IP 523	220°C

8. 15W40 Fully

Product type: Fully synthetic engine oil

Description: 15W-40 Fully Synthetic Engine Oil is engineered to deliver superior engine protection and consistent performance under extreme temperatures and heavy-duty conditions. Its advanced synthetic formulation ensures excellent thermal stability, strong wear protection, and reduced engine deposits.

Application: Designed for both diesel and petrol engines. It helps extend engine life, maintain cleanliness, and provide reliable lubrication during long drain intervals. Ideal for commercial vehicles, trucks, and high-load applications requiring dependable performance.

Product performance claim: API CK-4/SL.

Features and advantages:

- Strong wear protection
- Excellent thermal stability
- Cleaner engine performance
- Longer engine life
- Reliable heavy-duty performance

Typical technical results:

Parameter	Method	Typical Result
Viscosity Grade	SAE J300	15W40
Colour	VISUAL	Visual
Density @ 15°C	ASTM D4052	850.7 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	142.55 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	16.53 mm ² /s
Viscosity Index	ASTM D2270	160
Pour Point	ASTM D97	-30°C
Flash Point	IP 523	220°C

9. 15W40 SM/CF Mineral

Product type: Mineral engine oil

Description: A high-quality mineral engine oil formulated with refined base oils and a robust additive package. It provides strong lubrication, wear protection, and oxidation stability for gasoline and light-duty diesel engines. The formulation helps reduce deposits, minimize friction, and maintain smooth, reliable engine performance under various driving conditions.

Application: Suitable for gasoline and light-duty diesel engines used in passenger cars, light trucks, commercial vehicles, generators, and small industrial equipment. Recommended for mixed driving conditions such as high-temperature environments, frequent stop-and-go traffic, long-distance travel, and moderate load applications where dependable mineral-based protection is required.

Product performance claim: API SM/SL/SJ/CF; ACEA A3/B3 or A3/B4.

Features and advantages:

- Excellent protection against wear, deposits, and sludge
- Strong high-temperature stability and oxidation resistance
- Smooth engine operation with reduced friction
- Suitable for gasoline (SM) and older diesel engines (CF)
- Reliable performance in urban, highway, and light commercial use

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	866.8 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	137.92 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	15.796 mm ² /s
Viscosity Index	ASTM D2270	128.59
Pour Point	ASTM D97	-11°C
Flash Point	IP 523	240°C

10. 15W40 SQ/CF Semi

Product type: Semi-synthetic engine oil

Description: A high-performance semi-synthetic engine oil engineered for reliable protection and durability. It provides excellent lubrication under high temperature and heavy-load conditions, helping reduce engine wear and extend engine life.

Application: Suitable for a wide range of gasoline and diesel engines in passenger cars, light trucks, and commercial vehicles. Recommended for engines operating in mixed driving conditions, including city traffic and long-distance journeys.

Product performance claim: API SQ/CF-4; ACEA A3/B4; Cummins CES 20076/77; Mack EO-M Plus; Mercedes-Benz MB 228.3; MAN M3275; Volvo VDS-2.

Features and advantages:

- Superior wear and deposit protection
- Excellent oxidation stability at high temperatures
- Enhances engine performance and fuel efficiency
- Meets API SQ/CF performance standards
- Suitable for engines requiring semi-synthetic lubrication

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	875.6 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	135.17 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	15.938 mm ² /s
Viscosity Index	ASTM D2270	141.62
Pour Point	ASTM D97	-20°C
Flash Point	IP 523	240°C

11. 20W50 SL/CF Mineral

Product type: Mineral engine oil

Description: A high-quality mineral engine oil formulated with refined base oils and a balanced additive system. It provides excellent lubrication, wear protection, and high-temperature stability for reliable engine performance. The formulation helps reduce deposits, minimize friction, and maintain smooth operation under demanding conditions.

Application: Suitable for gasoline and light-duty diesel engines used in passenger cars, light trucks, motorcycles, generators, and commercial equipment. Recommended for engines operating in high-temperature environments, heavy-load driving, or long-distance travel where stable viscosity and dependable mineral-based protection are required.

Product performance claim: API SL/SJ/CF; ACEA A3/B3 or A3/B4.

Features and advantages:

- Excellent wear and deposit protection
- Stable viscosity and strong lubrication at high temperatures
- Reduced friction for smoother engine performance
- Suitable for gasoline (SL) and older diesel engines (CF)
- Reliable performance in city, highway, and heavy-load driving

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	868.1 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	188.07 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	19.996 mm ² /s
Viscosity Index	ASTM D2270	125.62
Pour Point	ASTM D97	-11°C
Flash Point	IP 523	240°C

12. ATF DEXRON-III

Product type: Automatic transmission fluid

Description: A premium automatic transmission fluid specially formulated for smooth shifting and reliable performance. It provides excellent oxidation stability, wear protection, and friction characteristics to extend transmission life.

Application: Suitable for automatic transmissions, power steering systems, and hydraulic systems requiring DEXRON-III specification. Ideal for cars, trucks, and other vehicles operating under normal or severe driving conditions.

Product performance claim: Toyota D-III; GM Dexron III F / Dexron III G; GM Type A Suffix A; Allison TES-389; Allison C-4; Mercedes Benz MB 236.1 / 236.9; MAN 339 Type V1 / V2, 339 Type Z1 / Z2; Ford Mercon IV; Ford M2C-138CJ; Voith H55.6335xx; Volvo STD 1273.41(97341); Subaru ATF / ATF HP; Mazda M III Fluid.

Features and advantages:

- Smooth gear shifting and reduced noise
- Protects against wear, corrosion, and foaming
- Maintains performance under high temperatures
- Compatible with systems requiring DEXRON-III, DEXRON-IIIE, and II specifications
- Enhances transmission life and reliability

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	857.2 kg/m ³
Kinematic Viscosity @ 40°C	ASTM D7042	37.078 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D7042	7.580 mm ² /s
Viscosity Index	ASTM D2270	178.53
Pour Point	ASTM D97	-31°C
Flash Point	IP 523	210°C

13. HD 40

Product type: Monograde diesel engine oil

Description: HD40 is a high-quality monograde diesel engine oil made from refined mineral base oils and selected additives. It provides effective protection against wear, deposits, rust, and oxidation, helping maintain reliable lubrication and extend engine life. The oil maintains good film strength and stability even under high operating temperatures.

Application: Recommended for naturally aspirated and light turbocharged diesel engines used in trucks, buses, construction, and agricultural equipment. Suitable for engines where SAE 40 monograde oil meeting API CF or SF / MIL-L performance levels is specified.

Product performance claim: API CF / SF; MIL-L-2104C; CCMC D2.

Features and advantages:

- Excellent wear protection for longer engine life
- High oxidation and thermal stability
- Protects against rust, corrosion, and sludge formation
- Maintains good oil film strength under high temperature
- Ensures smooth and reliable engine performance

Typical technical results:

Parameter	Method	Typical Result
Density @ 15°C	ASTM D4052	0.876 g/ml
Kinematic Viscosity @ 40°C	ASTM D445	101 mm ² /s
Kinematic Viscosity @ 100°C	ASTM D445	12.6 mm ² /s
Viscosity Index	ASTM D2270	120
Pour Point	ASTM D97	-10°C
Flash Point	ASTM D93	249°C

3. General Health and Safety Note

Based on the product data sheets, no adverse health effects are expected when the products are used for their intended applications and the recommendations in the Material Safety Data Sheet (MSDS) are followed. Products should not be used for purposes other than their intended use. When handling used products, care should be taken to protect health and safety.